

Assignment – 2.2

Q - 1. List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

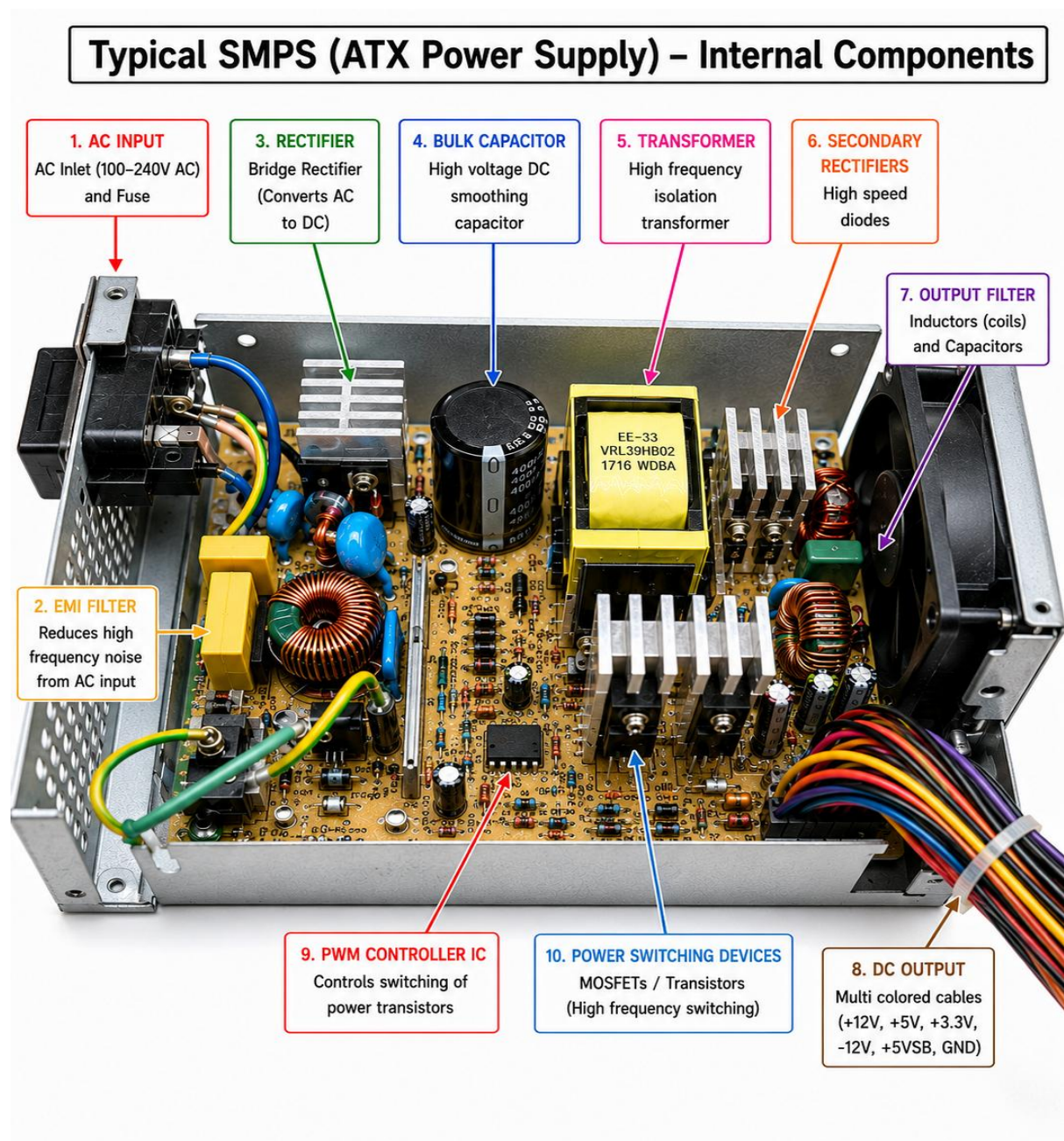
SMPS (Switched Mode Power Supply):

Device	Why it uses SMPS
Desktop Computer	SMPS converts AC power into the DC voltages needed by computer parts.
Television	SMPS provides the required low-voltage DC power to the TV circuits.
Mobile Charger	SMPS converts household AC into safe DC power for charging the phone.

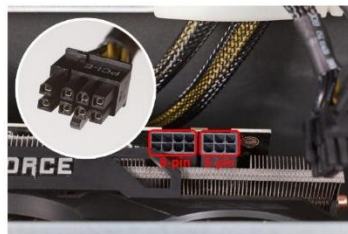
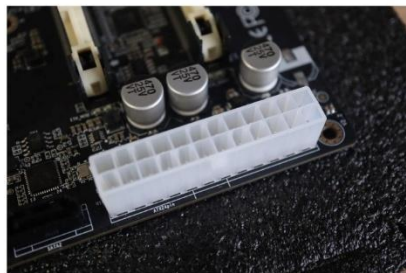
UPS (Uninterruptible Power Supply):

Device	Why it uses SMPS
Desktop Computer	UPS provides backup power during a power cut so work is not lost.
Wi-Fi Router	UPS keeps the internet connection running during short power failures.

Q - 2. Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.



Q - 3. Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.



Connector	Powers	Safety Tip
ATX 24-pin	Motherboard	Switch off PC before connecting.
SATA Power	HDD/SSD	Do not force the connector.
Molex	Old HDD, fans	Check direction before inserting.
PCIe 6/8-pin	Graphics card (GPU)	Use the correct PCIe cable.

Q - 4. Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

Steps to Connect SMPS in a Gaming PC

1. **Turn off and unplug** the PC before connecting cables.
2. **Connect the motherboard:** Plug the **24-pin ATX cable** into the motherboard's 24-pin power socket.
3. **Connect the CPU:** Plug the **4/8-pin EPS CPU cable** into the CPU power connector near the processor.
4. **Connect the graphics card:** Use the **PCIe 6/8-pin cable** from the SMPS and connect it to the GPU.
5. **Connect storage devices:** Use **SATA power cables** to power SATA SSDs or HDDs.
6. **Check connections:** Make sure all connectors are fully inserted and secure.
7. **Finish setup:** Connect the SMPS power cable to the wall socket, switch on the SMPS, and then press the PC power button.